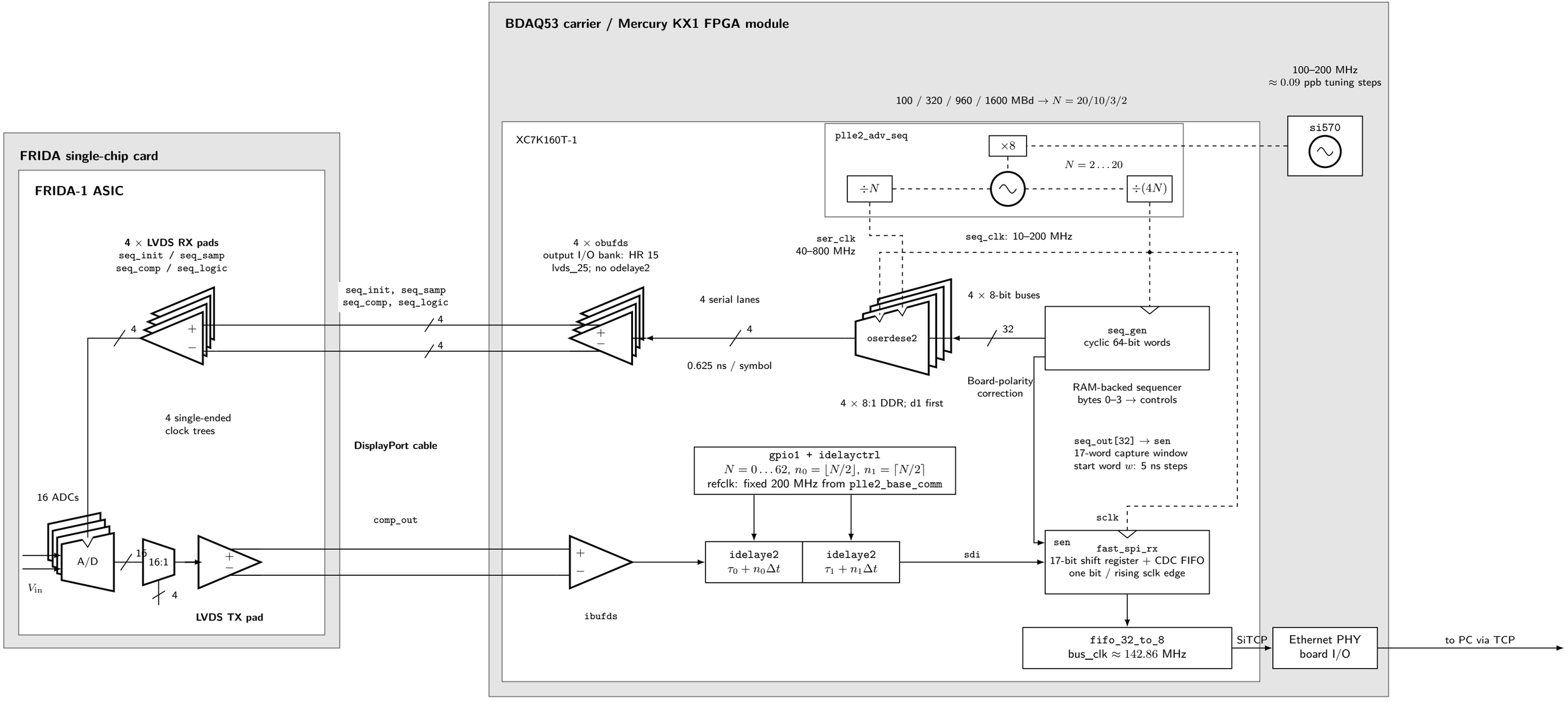

BDAQ53 carrier / Mercury KX1 FPGA module
100–200 MHz
≈ 0.09 ppb tuning steps
100 / 320 / 960 / 1600 MBd → N = 20/10/3/2
si570
XC7K160T-1
plle2_adv_seq
×8
N = 2 . . . 20
÷N
÷(4N)
FRIDA single-chip card
FRIDA-1 ASIC
4 × LVDS RX pads
seq_init / seq_samp
seq_comp / seq_logic
seq_init, seq_samp
seq_comp, seq_logic
4 × obufds
output I/O bank: HR 15
lvds_25; no odelaye2
ser_clk
40–800 MHz
seq_clk: 10–200 MHz
4 serial lanes
4 × 8-bit buses
oserdese2
seq_gen
cyclic 64-bit words
0.625 ns / symbol
Board-polarity
correction
RAM-backed sequencer
bytes 0–3 → controls
4 × 8:1 DDR; d1 first
4 single-ended
clock trees
DisplayPort cable
seq_out[32] → sen
17-word capture window
start word w: 5 ns steps
gpio1 + idelayctrl
N = 0 . . . 62, n0 = ⌊N/2⌋, n1 = ⌈N/2⌉
refclk: fixed 200 MHz from plle2_base_comm
16 ADCs
comp_out
sclk
sen
fast_spi_rx
17-bit shift register + CDC FIFO
one bit / rising sclk edge
A/D
16:1
idelaye2
τ0 + n0Δt
idelaye2
τ1 + n1Δt
sdi
Vin
ibufds
LVDS TX pad
fifo_32_to_8
bus_clk ≈ 142.86 MHz
SiTCP
Ethernet PHY
board I/O
to PC via TCP